CSIT321 PROJECT DESCRIPTIONS
APRIL 2026 – SEPTEMBER 2026

Projects available:

CSIT-26-S2-01	**Gaming with Bare Hands: A Machine Learning Approach**
CSIT-26-S2-02	**TripAid based on LOLs**
CSIT-26-S2-03	**Counting Maize Tassels in the Wild via Deep Neural Network**
CSIT-26-S2-04	**Smart Task Allocation Application**
CSIT-26-S2-05	**A Mobile Application for Wise Workout**
CSIT-26-S2-06	**Linkable Ring Signatures and Applications**
CSIT-26-S2-07	**Homomorphic Encryption and Its applications to e-voting**
CSIT-26-S2-08	**A secure file sharing and recovery software system**
CSIT-26-S2-09	**Private Set Intersection for Healthcare Application**
CSIT-26-S2-10	**Aggregate Signatures for IoT**
CSIT-26-S2-11	**Seamless File Encryption App for Cloud**
CSIT-26-S2-12	**Implementation of online auction platform**
CSIT-26-S2-13	**Webservice for Stock Market Prediction**
CSIT-26-S2-14	**Pregnancy and Mobile App**
CSIT-26-S2-15	**Social network analysis platform**
CSIT-26-S2-16	**Chatbot for Customer Support**
CSIT-26-S2-17	**Face Recognition: A Deep Learning Approach**
CSIT-26-S2-18	**A Mobile Application for Food Diary and Blood Glucose Record**
CSIT-26-S2-19	**SENSE Spaces Planning Tool**
CSIT-26-S2-20	**Safe QR**
CSIT-26-S2-21	**DeepPurple**
CSIT-26-S2-22	**Client-Side Script Security Inspector**
CSIT-26-S2-23	**Rockland**
CSIT-26-S2-24	**Big data-driven video recommendation system**

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Project ID	CSIT-26-S2-01
Proposed Title	**Gaming with Bare Hands: A Machine Learning Approach**
Project Description	Machine learning algorithms have recently been developed to enable computer to reliably recognise hand gestures. This provides the opportunity to play computer games with bare hands. This project will investigate existing machine learning based hand gesture recognition algorithms and improve it to effectively play a computer game such a racing car, aviation, or the alike. This project will contribute to human-machine interaction, computer vision, and computer games and demonstrate the capability of modern AI techniques. This project is related to Deep Learning, Neural Networks, Image Processing, and Programming.

Project ID	CSIT-26-S2-02
Proposed Title	**TripAid based on LOLs**
Project Description	This project aims to develop a native iOS and Android app to allow travellers to move around Singapore with ease, providing curated walking routes, points of interest and written guides in one nifty app. A major component is a forum section to provide a social aspect to the user experience, as well as a place to ask questions about destinations and find answers from fellow travellers. Additionally, the app should be able to integrate with Local Opinion Leaders (LOLs), or influencers, when going to market. This may be through the forum, for example posting questions specifically for the influencer. The influencer will be a resident in Singapore who will have a large following on social media platforms. It would be good if the app can integrate with video-sharing platforms such as TikTok. The project always looking new functions that will disrupt the way things are currently done. While working on the project the team needs to look at travel guides, travel booking sites etc that are used by travellers to plan and book overseas travel, and then come up with ideas about how to disrupt the way things or currently done, or any new and interesting functionality that would make the app very compelling to users.

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Project ID	CSIT-26-S2-03
Proposed Title	Counting Maize Tassels in the Wild via Deep Neural Network
Project Description	Problem description: Over the years, emerging interest has occurred in integrating computer vision technology into agriculture industry. In this project, we sill study one critical problem in this area: counting maize tassels in the wild. Accurate counting of maize tassels is important for monitoring the growth status of maize plants. But this tedious work is traditionally done via manual work. To free humans from such intensive work and increase the scalability required for plant analysis, in this project we aim to develop robust deep neural network to automatically count the maize tassels. The expected outcome will be a system that can automatically read maize plant image, predict the maize tassel number, and visualize the counting result. Expected benefits: The developed algorithm is expected to be able to automatically count maize tassels in the wild with acceptable accuracy. This will largely reduce the human labour and meet the need of large-scale analysis of phenotype in agriculture.

Project ID	CSIT-26-S2-04
Proposed Title	Smart Task Allocation Application
Project Description	A small company has an issue with managing task allocation for casual employees. Currently, the eligibility and working time of the casual staff are manually managed by one manager. Whenever, other department want to allocate tasks to any of the casual staff, they need to update the manager who will check the eligibility of the nominated staff. The process takes time and may have human-errors. The company requests an application that could help to automate the management. The application should allow staff from related department to allocate working time to any casual staff in the list. Then, the working time of the casual staff will be automatically updated. So, we can track the time eligibility of the casual staff. This is a small company so open-source software would be preferable.

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Project ID	**CSIT-26-S2-05**
Proposed Title	**A Mobile Application for Wise Workout**
Project Description	This project aims to develop a mobile application (for Android, iOS, or both) that can collect exercise data from cellphone sensors or wearable devices, provide training and fitness advice, and facilitate social interactions such as initiating competitions between registered users. The application should have the following features. 1. Collect exercise data: The application can collect user exercise information from either the cellphone's own sensors or from smart wearable devices (via Wi-Fi, Bluetooth, etc.). 2. Estimate exercise effects (short-term or long-term) and provide analysis: The application can estimate the exercise results of each activity, a day, a week, a month, or any period, such as the number of calories burned and the average and maximum heart rate. 3. Supply fitness advice and schedule a fitness plan: The application can provide fitness advice according to the user's fitness goals and their entered health information. It can also create a fitness plan based on the fitness goal, and users can customize the plan to their preferences, such as the length of the plan or the daily weight loss goal. 4. Remind user of exercise or take a break: The application can send reminders to users to exercise if they have not met their scheduled exercise goals. It can also send reminders to users to take a break if they have been exercising too much. 5. Connect with social media and initiate competitions between users: The application can be used to share exercise completion data on social media platforms such as Facebook, Instagram, and Twitter. Users can also initiate competitions with other registered users and invite them to join in. The competition results and rankings can be shared on social media. The project may consider releasing the application to the app store.

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Project ID	**CSIT-26-S2-06**
Proposed Title	**Linkable Ring Signatures and Applications**
Project Description	Ring signature [1] is a type of digital signature that allows a signer to sign a message on the behalf of a group that he is belong to such that the resulting signature does not reveal his identity. Formation of a group is spontaneous: diversion group members can be totally unaware of being conscripted to the group. Linkable ring signature [2] is a variant of ring signature but with added linkability: such signatures allow anyone to determine if they are signed by the same group member (i.e. they are linked). If a user signs only once on behalf of a group, he still enjoys anonymity similar to that in conventional ring signature schemes. If the user signs multiple times, anyone can tell that these signatures have been generated by the same group member. Linkable ring signatures have many applications, especially in e-voting and blockchain. The linkable ring signature method is used within the Monero cryptocurrency and defined by the Ring Confidential Transaction (RingCT) [3]. This project is to study linkable ring signatures and the efficient RingCT construction [3,4]. The students are also required to provide an efficient implementation for such RingCT. **References:** [1] Ronald L. Rivest, Adi Shamir, Yael Tauman: How to Leak a Secret. ASIACRYPT 2001: 552-565 [2] Joseph K. Liu, Victor K. Wei, Duncan S. Wong: Linkable Spontaneous Anonymous Group Signature for Ad Hoc Groups (Extended Abstract). ACISP 2004: 325-335 [3] Shen Noether: Ring SIgnature Confidential Transactions for Monero. IACR Cryptol. ePrint Arch. 2015: 1098 (2015) [4] Tsz Hon Yuen, Shifeng Sun, Joseph K. Liu, Man Ho Au, Muhammed F. Esgin, Qingzhao Zhang, Dawu Gu: RingCT 3.0 for Blockchain Confidential Transaction: Shorter Size and Stronger Security. Financial Cryptography 2020: 464-483

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Project ID	**CSIT-26-S2-07**
Proposed Title	**Homomorphic Encryption and Its applications to e-voting**
Project Description	Homomorphic encryption is a form of encryption that permits users to perform computations on its encrypted data without first decrypting it. These resulting computations are left in an encrypted form which, when decrypted, result in an identical output to that produced had the operations been performed on the unencrypted data. Homomorphic encryption can be used for privacy-preserving outsourced storage and computation. This allows data to be encrypted and out-sourced to commercial cloud environments for processing, all while encrypted. Fully homomorphic encryption (FHE) allows the evaluation of arbitrary circuits composed of multiple types of gates of unbounded depth, and is the strongest notion of homomorphic encryption. It was raised [1] in 1978 but the first FHE scheme was proposed by Gentry in his PhD thesis in 2009 [2] and which was later extended and improved by many authors. FHE has been becoming an important research direction and has many applications in cloud computation, privacy-preserving machine learning, e-voting, etc. In this project, we will investigate the FHE scheme from integers [3] and its applications in e-voting [4]. The students are also required to implement the resulting e-voting system. **References:** [1] R. L. Rivest, L. Adleman, and M. L. Dertouzos. On data banks and privacy homomorphisms. In Foundations of Secure Computation, 1978. [2] C. gentry. A Fully Homomorphic Encryption Scheme. PhD thesis, Stanford 2009. [3] Marten van Dijk, Craig Gentry, Shai Halevi, Vinod Vaikuntanathan: Fully Homomorphic Encryption over the Integers. EUROCRYPT 2010: 26-43 [4] Xun Yi, Russell Paulet, Elisa Bertino: Homomorphic Encryption and Applications. Springer Briefs in Computer Science, Springer 2014, ISBN 978-3-319-12228-1, pp. 1-126

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Project ID	**CSIT-26-S2-08**
Proposed Title	**A secure file sharing and recovery software system**
Project Description	This project aims to develop and implement a secure file sharing and reconstruction software system. The software allows a user to split and store a secret file in different locations/devices so that when a minimum number of fragments located at specified locations are gathered, original file can be reconstructed. This system will provide better security than encryption based approaches, as well as fault tolerance in the case some storage devices are lost/damaged. The software system should have a GUI to allow a user to perform file sharing/reconstruction operations. The operations could be password protected. The system can be built based on a secret sharing scheme or error correction code.

Project ID	**CSIT-26-S2-09**
Proposed Title	**Private Set Intersection for Healthcare Application**
Project Description	This project is to develop a system for applying the Private Set Intersection (PSI) protocol for a healthcare application. Simply speaking, the PSI protocol allows two parties, Alice and Bob, having the sets X and Y, respectively, to calculate $X \cap Y$, without revealing the members of X and Y to each other. In the scenario for this project, suppose that a patient sequenced her DNA and obtained a set of genes $X = \{x_1, x_2, ,,, x_m\}$. The patient wants to find the risk of having a particular disease in the future. To achieve this, the patient will use the service of a specialized hospital, which discovered a set of genes, $Y=\{y_1, y_2, \ldots, y_n\}$, that increase the risk of contracting the disease. We assume that having the gene y_i in the patient's sequence increases the risk by a constant c%. The challenge here is the patient does not want to reveal to the hospital her set of genes (because it is a personal and highly sensitive piece of information) and the hospital does not want to reveal its discoveries (i.e., the set of genes that cause the disease). To address the challenge, the project uses the Diffie-Hellman based PSI* to give the patient the risk that she could have the disease while she does not reveal her genes to the hospital and the hospital does not reveal its discoveries either. Note that the risk here is calculated as $c\|X \cap Y\|/n$ %. * Huberman B A, Franklin M, Hogg T. Enhancing privacy and trust in electronic communities, ACM Conference on Electronic Commerce. ACM, 1999:78-86

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Project ID	**CSIT-26-S2-10**
Proposed Title	**Aggregate Signatures for IoT**
Project Description	The "Internet of Things" (IoT) has many applications, from household appliances like coffee machines and toasters to medical devices like a patient's heart-rate monitor. As with any other network environment, security is always a concern for the IoT environment. This project is to implement Gentry's identity-based aggregate signature (IBAS)* scheme for the IoT environment where devices are represented by Raspberry Pi (or similar devices.) The IBAS scheme provides the ability to aggregate multiple identity-based signatures and verify them as one. Note that in an identity-based scheme, the signer's identifier (e.g., hardware address or serial number), instead of a digital certificate, can be used to verify a signature. More specifically, the project will show how the implemented IBAS scheme can be applied to aggregate signatures from various weather data updates generated by multiple IoT devices and authenticate them. The project requires to produce a feasible API that can be applied to a wireless IoT network. * Craig Gentry and Zulfikar Ramzan. Identity-Based Aggregate Signatures. In: Yung, M., Dodis, Y., Kiayias, A., Malkin, T. (eds) Public Key Cryptography - PKC 2006. PKC 2006. Lecture Notes in Computer Science, vol 3958. Springer, 2006

Project ID	**CSIT-26-S2-11**
Proposed Title	**Seamless File Encryption App for Cloud**
Project Description	Although there are numerous services that cloud computing can provide, storing files on the cloud can be the most common service for all levels of users. However, there is an obvious privacy issue in that files are stored in the cloud in plaintext unless users encrypt them. Encrypting files before uploading them to the cloud is easier said than done. The project aims to develop a client app for encrypting and decrypting files when the user uploads and downloads them, respectively, from any cloud service. The application should use a state-of-the-art authenticated encryption mode, such as GCM or CCM, to encrypt files. The application should be seamless and user-friendly in the sense that once the user has created a key for encrypting files, she doesn't have to do anything other than select files to upload and hit the "upload" button. The decryption should also be seamless, as the user must select a file to download and hit the "download" button. The application should also support key management and updates.

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Project ID	CSIT-26-S2-12
Proposed Title	Implementation of online auction platform
Project Description	With the development of Web technology and the prevalence of online activities, e-commerce has become more and more prosperous. An online auction system is based on the C2C e-commerce model which has great advantages over traditional auctions, e.g., eliminate the need for intermediaries. This project focuses on design and implementation of an online auction platform. The platform can use JSP + Servlet + JavaBean development design pattern following the MVC three-tier architecture model. The platform should integrate the features such as 'price up', 'low start high', and 'public bidding'. This platform can utilize user-based collaborative filtering algorithms to recommend goods for users according to their retrieval or bidding habits. This enables users to bid on commodities conveniently on the auction platform, as well as to utilize their unused items to participate in the bidding.

Project ID	CSIT-26-S2-13
Proposed Title	Webservice for Stock Market Prediction
Project Description	This project aims to implements a webservice that provides stock market forecasting and recommendation. This webservice should at least include two components, i.e. a frontend application and a backend engine. The frontend application accepts real-time the stock market data (e.g., via RESTful protocol or a streaming platform) and provides forecasting service to users. The backend engine uses machine learning to predict the stock trends and provide recommendation. In addition, the machine learning models must be updatable with the up-to-date stock market data.

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Project ID	CSIT-26-S2-14
Proposed Title	Pregnancy and Mobile App
Project Description	Pregnancies necessarily need knowledge about nutrition and special care for their before-during-after pregnant periods. However, not all women can afford the services and they may miss important care for them as well as their baby during the periods. Therefore, we would like to create a mobile application that should include the essential resources for them. Additionally, specialists can provide free consultation for the women. The mobile application should be free for everyone. It should be user friendly for both pregnancy and doctors to communicate and share knowledge. This app should allow both pregnancy and doctors to create their profile as well as services that they need or can provide, respectively. The app should present a list of categories of useful materials/lessons/advice for pregnancies. It can also support them with consultation with specialists who are volunteers. The application can enable volunteers to advertise their services there and match the advertisements with requested from pregnancies.

Project ID	CSIT-26-S2-15
Proposed Title	Social network analysis platform
Project Description	The Social Network Analysis project involves utilizing data science and network analysis techniques to study and extract insights from social networks. The project aims to understand the structure, relationships, and dynamics within a social network, providing valuable insights for businesses, researchers, and individuals. Key Tasks: 1. Gather data from social media platforms, online communities, or any relevant sources. Clean and preprocess the data to remove noise and ensure data quality. 2. Network Visualization: Visualize the social network graphically to provide an overview of its structure. Utilize visualization tools and libraries to create informative and interactive network visualizations. 3. Centrality and Influence Analysis: Calculate centrality measures (e.g., degree, betweenness, closeness) to identify influential nodes within the network. Analyze individuals/entities that play critical roles in connecting others. 4. Predictive Analysis: Develop predictive models to forecast future network growth, behavior, or trends. Utilize machine learning techniques to make predictions based on historical data.

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Project ID	**CSIT-26-S2-16**
Proposed Title	**Chatbot for Customer Support**
Project Description	The Chatbot for Customer Support project involves the development of an AI-powered chatbot that assists customers in real-time, answering their inquiries and providing solutions to common issues. The chatbot acts as a virtual customer support agent, enhancing user experience and reducing the workload on human agents. The project focuses on leveraging natural language processing (NLP) techniques and machine learning algorithms to create an efficient and responsive chatbot system. Key Tasks: 1. Requirement Analysis: gather detailed requirements from stakeholders to understand the scope and objectives of the chatbot. 2. NLP Framework Selection: Research and select a suitable NLP framework or library for chatbot development, such as Dialogflow, Rasa, or Microsoft Bot Framework. 3. Response Generation: Design response templates for different intents and entities. Develop algorithms for generating contextually relevant responses based on user queries. 4. User Interface Design: Design a user-friendly chat interface with appropriate styling and branding. Incorporate interactive elements like buttons and quick replies to guide users through interactions.

Project ID	**CSIT-26-S2-17**
Proposed Title	**Face Recognition: A Deep Learning Approach**
Project Description	Given a still facial image of a person, this project aims to correctly identify the person due to its specific facial feature. This has recently become an active research topic with the technique of generative adversarial networks (GANs). This project will investigate existing GANs or other deep learning based models for this task and develop an improved model for it. This project is related to Deep Learning, Neural Networks, Image Processing, and Programming.

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Project ID	**CSIT-26-S2-18**
Proposed Title	**A Mobile Application for Food Diary and Blood Glucose Record**
Project Description	This project is to develop a mobile application (Android, iOS or both) to collect record daily diet and every blood sugar reading. The application should include the following functions. 1. Record food consumption and compute calories intake: The application can take user inputs on the type and quantity of food and calculate how many calories intake of every meal and every day. 2. Support barcode or physical object scanning: The application can identify the food and estimate the unit calorie by scanning either the food itself or the barcode on the packaging bag. Thereafter, the user only needs to manually enter the consumed quantity to learn the energy s/he has eaten. 3. Track blood glucose readings: The application can take user inputs on blood sugar levels and measurement time. The application can also read sugar level and time from the glucose test meter via wireless way, say, Wi-Fi, Bluetooth, etc., or by taking a photo of the meter screen. 4. Graph energy intake and glucose levels: The application can derive the statistics and analysis for a period of user interest, showing the correlation between calorie intake and blood sugar levels. 5. Predict blood glucose levels: After learning each pair of calorie intake and glucose readings, the application can compute or adjust the estimated user's basal metabolism. Thereafter, the user enters all food in a meal, the application can estimate the glucose level at the desired time so that the user may adjust the energy intake in advance. Advanced features may include the following. 1. Consider the influence of injected insulin on glucose level prediction in addition to calorie intake and basal metabolism. 2. Share data with healthcare providers. 3. Set reminders for blood sugar testing.

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Project ID	CSIT-26-S2-19
Proposed Title	SENSE Spaces Planning Tool
Project Description	The SENSE Spaces Planning Tool is a web application which allows its users to plan and visualise their own multi-sensory environments. This application renders a 3D room based on dimensions provided by the user, and allows them to add, subtract and move around (3D rendered) multi-sensory objects within the space. The user is able to customise their view of the room by using the on-screen controls to zoom and rotate the screen. Objects are found in the right-hand menu and are categorised by sense. Users can search for objects in this menu and can also view additional object information. In the left-hand menu, users are able to create a new room, import an existing room, export information about the space, and also save their personalised user preferences. The target audience for SENSE Spaces Planning Tool includes: people with sensory processing differences, people accessing mental support, those seeking a therapeutic environment, educators, parents, support workers, and WHS staff. Technologies used in the development of the SENSE Spaces Planning Tool include: HTML, CSS, JavaScript, and Three.js

Project ID	CSIT-26-S2-20
Proposed Title	Safe QR
Project Description	QR code is a popular information-sharing method for text, images, audio, URL, etc. However, users are undertaking potential issues of being attacked when scanning a QR code. For example, a malicious QR code may contain a web link that may steal personal information when the webpage is loaded. Shared content, such as images and audio, may be damaged or have malicious code. It is difficult for users to detect and react within a limited time. Safe QR is a project that aims to develop a mobile application to enhance data security when using QR code. The key functionalities should include: • Read QR code from a camera or a picture. • Detect insecure links from the QR code. • Open links in a protected environment, e.g., sandboxing. This may integrate the existing component from the browsers like Chrome. • QR code payload inspection. This will detect potential issues in the text content, such as malicious code to other apps, phishing SMS or phone numbers, etc. This project should develop a real-world application to demonstrate the advantages of Safe QR or detect unsafe QR code.

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Project ID	**CSIT-26-S2-21**
Proposed Title	**DeepPurple**
Project Description	Effective communication is about more than just exchanging information. It's about understanding the emotions and intentions behind the words. DeepPurple is the solution. It's a text analysis software that automates the process of looking at the communications clients write and deciphers the emotions they express, to get a deep understanding of what they're saying and how they feel. Consider this process scaled to millions of communications and customers. Through the use of an AI engine, this natural language analysis software mines emotion, what these emotions relate to, and displays the insights in an interactive and adaptive UI. Companies will be empowered to better understand their clients and customise their approach to every aspect of the business. Furthermore, DeepPurple provides a comprehensive understanding of both detailed and general trends over time. We turn words into insights. • Technologies used: OpenAI GPT3 • React • Spring Boot • AWS Lambda • AWS Elastic Beanstalk • AWS RDS • AWS Amplify

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Project ID	CSIT-26-S2-22
Proposed Title	Client-Side Script Security Inspector
Project Description	Many web security issues are caused by client-side script vulnerabilities. For instance, a web page may apply a compromised JavaScript library that contains polluted prototypes; a login page does not implement proper prevention techniques against client-side Cross-Site Scripting attacks; an HTTP header does not well configure essential security properties. These issues are detectable by inspecting the html page source code and the headers information. This project aims to develop a browser extension that can review the script of a loaded html page (or if possible before the page is loaded) to check the critical security indicators, such as the source of the JavaScript library, CSRF token implementation, and user input sensitization. This extension should give a report about the security risks of the viewing page based on the client's point of view. This task may not be easy because the needed secure mechanisms will vary on different pages. The group should develop their knowledge of fundamental client-side web security and prepare a list of security indicators for review. This project should also look after user privacy that the extension cannot leak the user information or the security report.

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Project ID	CSIT-26-S2-23
Proposed Title	Rockland
Project Description	Rocks convey secrets of the earth. By discovering different types of rocks, we can learn about the environment, history, and future. However, few applications can provide interactive education about rocks. It is needed to develop an easy-to-use App that can show rock distribution on the map and provide relevant information when we click on it. The key functionalities are designed as follows. • A map-based rock distribution that can show the relevant rock information. • Allows users to upload/edit information/comments about the rocks. • Virtually collect rocks via the App. • Game-like rock discovery in the surrounding area. • Recognise rocks by using machine learning technologies. • Provide incentives to retain the users. This project aims to develop an application for learning rocks. It is expected to • Provide an interactive and easy-to-use App to deliver knowledge about rocks. • Help with rock collection and education from nature. • Encourage people to learn our environment. The group need to identify target customers and develop suitable incentive scheme for attracting and keeping users.

Project ID	CSIT-26-S2-26
Proposed Title	Big data-driven video recommendation system
Project Description	The objective of this project is to develop a video recommendation system that incorporates advanced collaborative filtering algorithms. The system should: 1. Implement collaborative filtering algorithms to enhance the accuracy of content recommendations. 2. Develop a real-time recommendation component for timely content suggestions. 3. Develop an offline recommendation mechanism to ensure recommendations remain relevant. 4. Integrate the Spark platform to enable efficient big data processing and recommendation system implementation. 5. Provide detailed insights into the overall framework design, module interactions, recommendation algorithm design, and database architecture.

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